

Crypto Quant Research Report

This report summarizes a systematic crypto trading research project using the TWSQ Trading Library. The project evaluates momentum, breadth-weighted momentum, capitulation-reversal strategies, and combined portfolios under realistic transaction costs.

Research Questions

1. Can cross-sectional crypto momentum outperform BTC buy-and-hold?
2. Can a low-correlation capitulation strategy improve portfolio Sharpe?
3. Are strategy returns explained by BTC beta or by independent alpha?

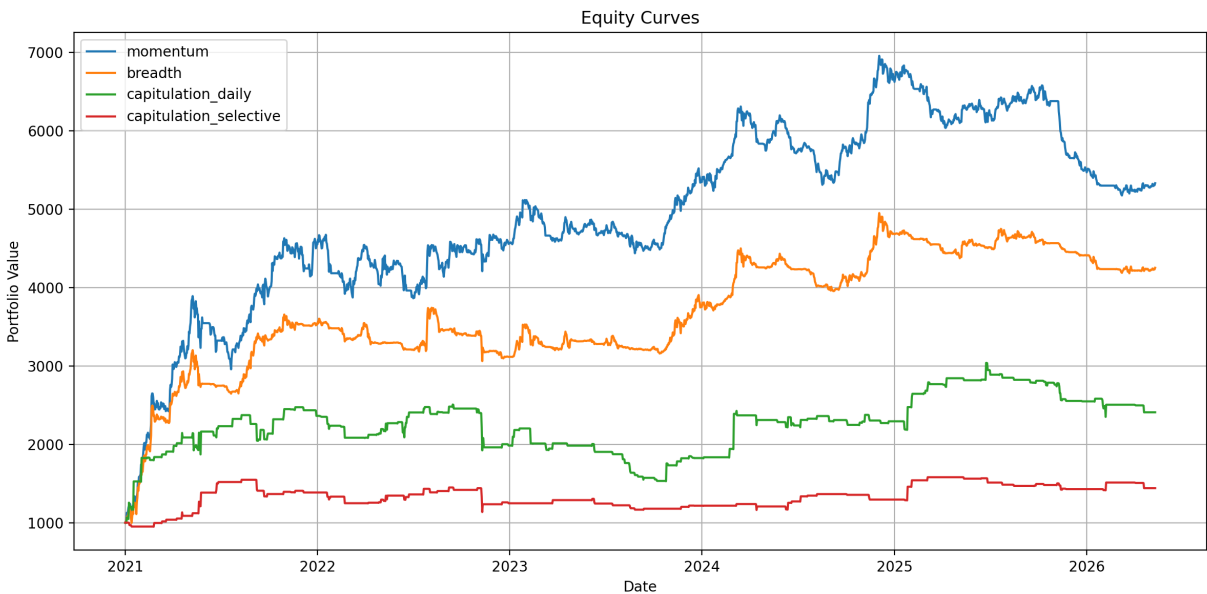
Strategies Evaluated

- 1. Momentum Strategy:** Selects the most liquid cryptocurrencies and ranks them using a volatility-adjusted moving average signal. The strategy buys assets with positive trend scores.
- 2. Breadth-Weighted Momentum:** Extends the momentum strategy by scaling exposure according to market breadth. If many coins are in positive trends, the strategy invests more capital; if market breadth is weak, it reduces exposure.
- 3. Capitulation Bounce Daily:** A reversal strategy that buys liquid assets after large negative daily returns combined with abnormal volume spikes, expecting short-term panic reversals.
- 4. Capitulation Bounce Selective:** A more conservative version of the capitulation strategy with stricter entry filters and lower exposure. It trades less frequently and aims to reduce drawdowns.

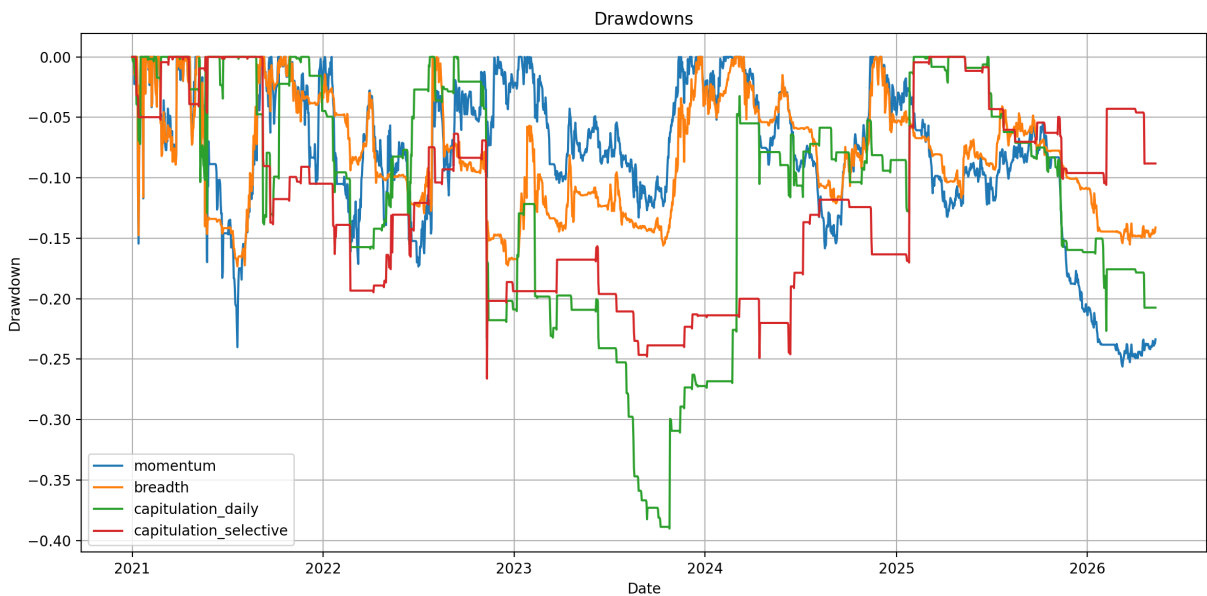
Standalone Strategy Summary

strategy	final_value	profit	total_return	sharpe	max_drawdown	turnover_multiple	num_orders
momentum	5329.4335	4329.4335	4.3487	1.2840	-0.2562	452.5186	5377.0000
breadth	4250.2617	3250.2617	3.2595	1.2620	-0.1819	275.1921	5377.0000
capitulation_daily	2407.3115	1407.3115	1.4073	0.6733	-0.3902	395.4043	664.0000
capitulation_selective	1439.9659	439.9659	0.4400	0.4610	-0.2662	114.5814	270.0000

Equity Curves



Drawdowns



Strategy Correlations

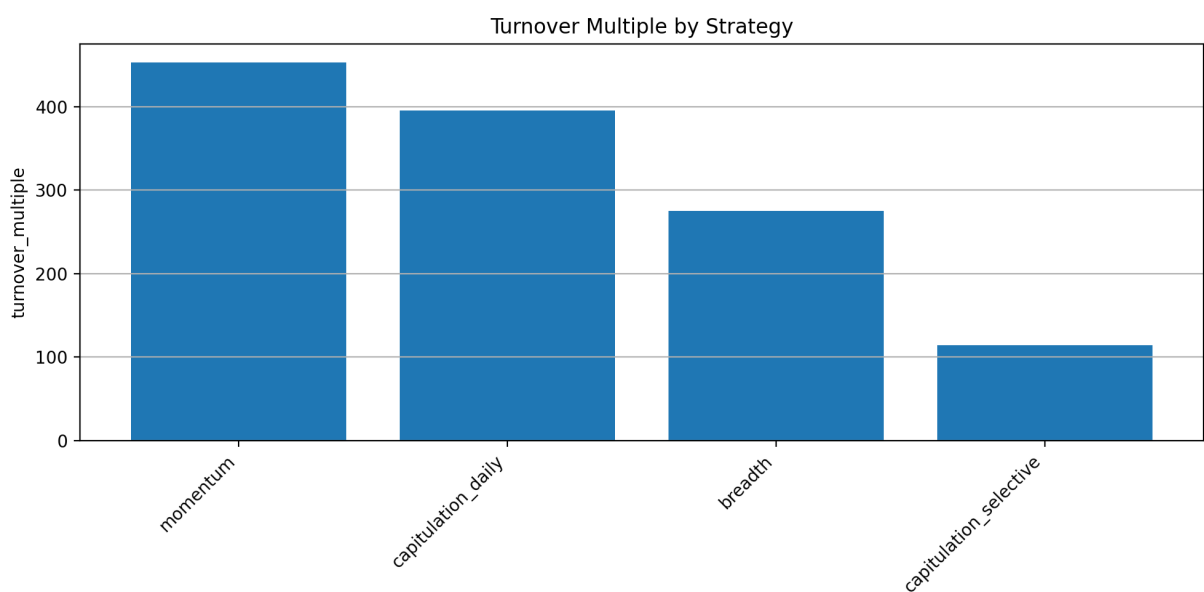
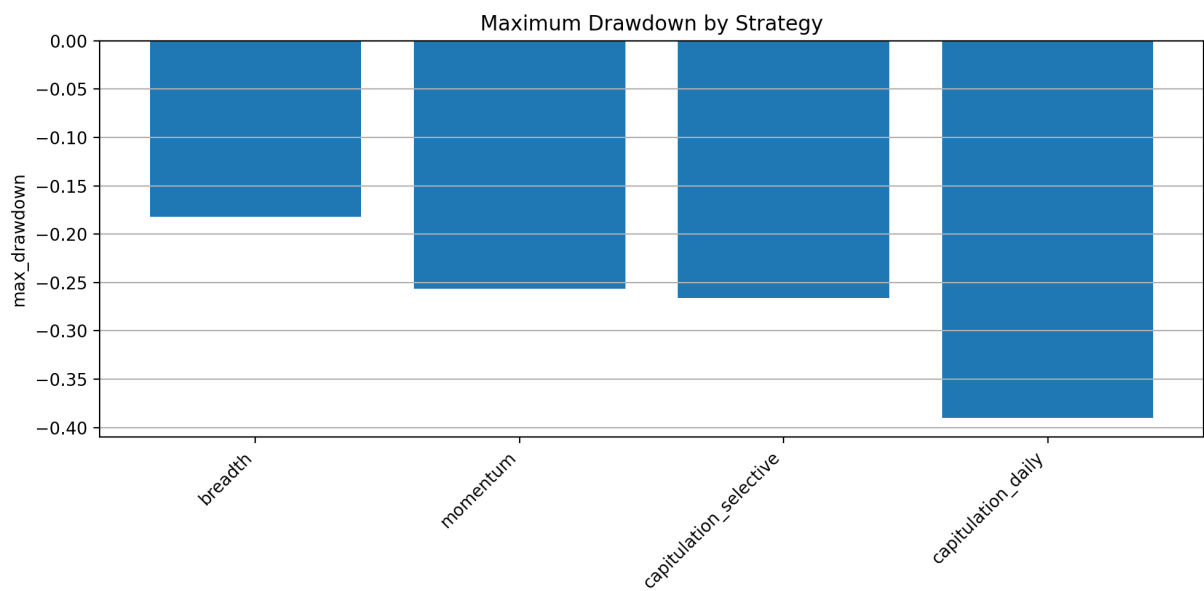
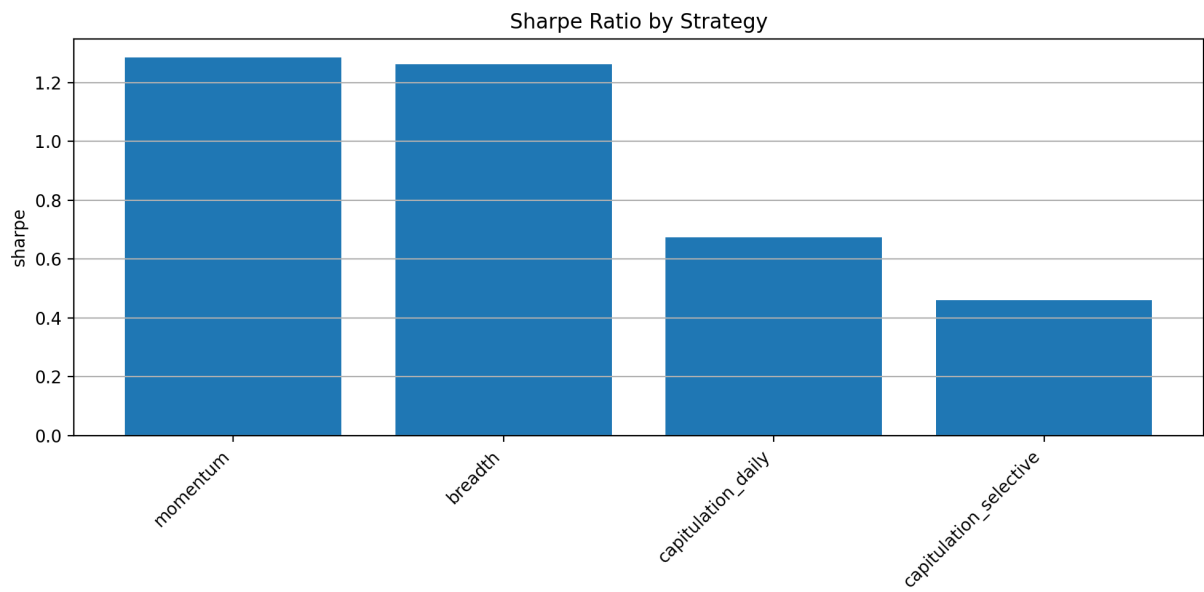
Low correlation between momentum and capitulation strategies suggests that capitulation can act as a diversifying alpha source.

Correlation Matrix

Unnamed: 0	momentum	breadth	capitulation_daily	capitulation_selective
momentum	1.0000	0.8886	0.2697	0.2064
breadth	0.8886	1.0000	0.2394	0.1652
capitulation_daily	0.2697	0.2394	1.0000	0.5506
capitulation_selective	0.2064	0.1652	0.5506	1.0000



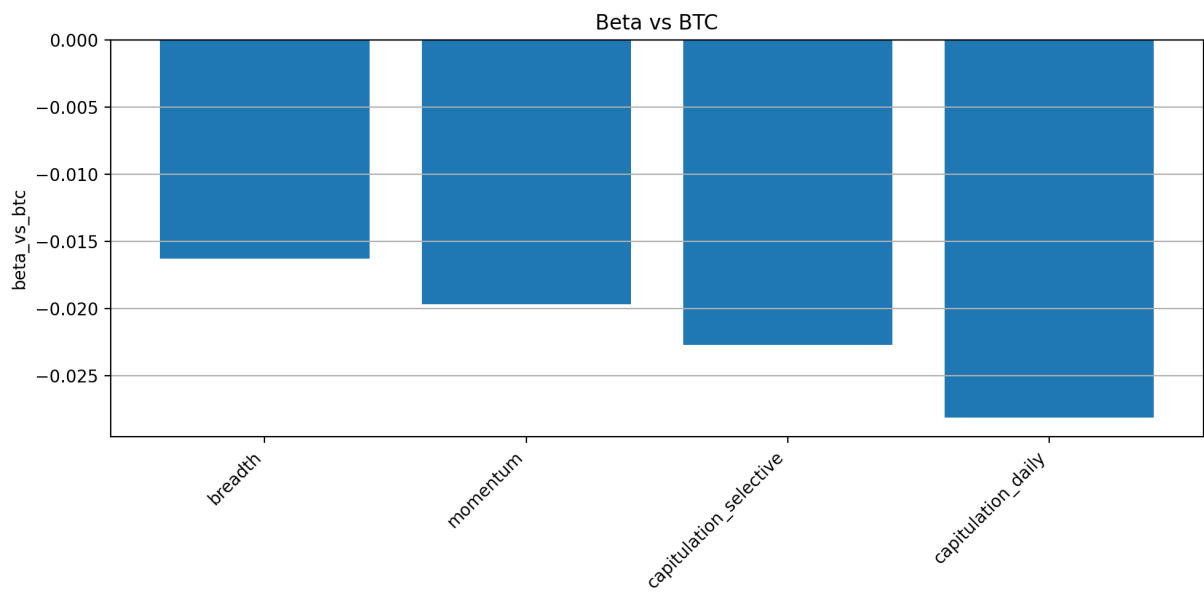
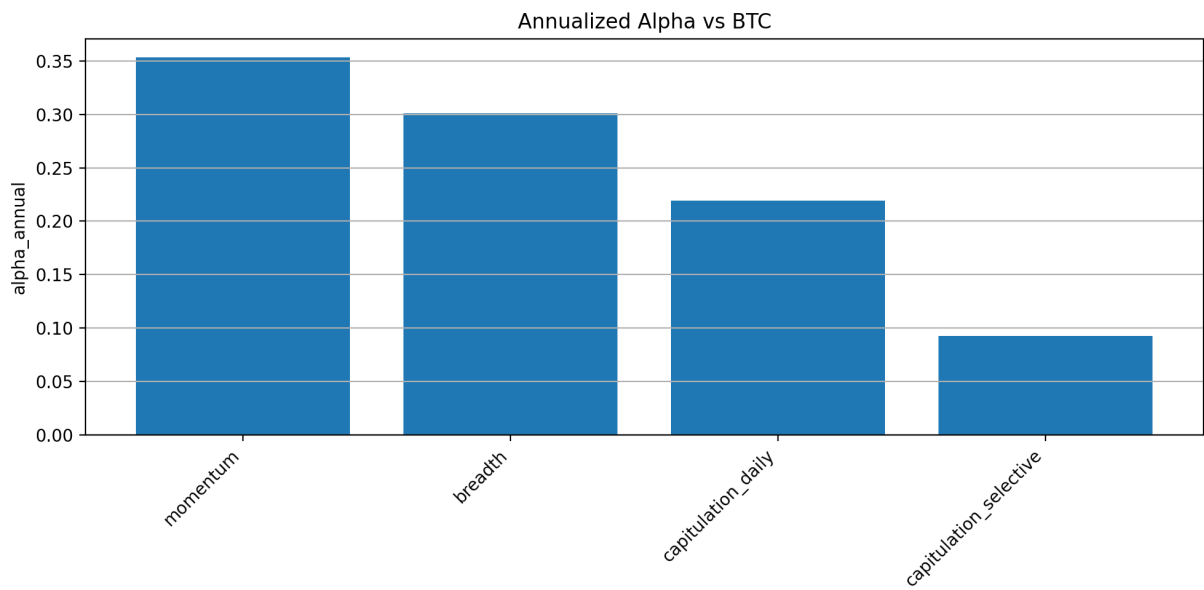
Performance Metrics



Alpha/Beta Regression vs BTC

strategy	alpha_daily	alpha_annual	beta_vs_btc	r_squared	information_ratio	alpha_t_stat	alpha_p_value
momentum	0.0010	0.3532	-0.0197	0.0018	1.2989	3.0031	0.0027
breadth	0.0008	0.3013	-0.0163	0.0016	1.2777	2.9541	0.0032
capitulation_daily	0.0006	0.2190	-0.0281	0.0028	0.7081	1.6371	0.1018
capitulation_selective	0.0003	0.0925	-0.0227	0.0052	0.5051	1.1678	0.2430

The alpha/beta regression estimates whether the strategy returns are explained by BTC buy-and-hold exposure. Low beta and low R-squared indicate that strategy returns are not simply crypto market beta.



Parameter Robustness

To reduce the risk of overfitting, the main strategy parameters were varied around the baseline values. The goal was not to optimize parameters, but to verify whether the strategy remains profitable across reasonable nearby configurations.

Momentum Parameter Robustness - Top 10 by Sharpe

final_value	profit	total_return	sharpe	max_drawdown	turnover_multiple	num_orders	fast_window	slow_window	n_winners
5429.8712	4429.8712	4.4495	1.4308	-0.2509	386.3249	8139.0000	15.0000	30.0000	5.0000
6193.2508	5193.2508	5.2156	1.2979	-0.2390	401.5142	3699.0000	10.0000	50.0000	2.0000
5852.6350	4852.6350	4.8738	1.2965	-0.3582	527.5483	5477.0000	5.0000	30.0000	3.0000
5700.0337	4700.0337	4.7206	1.2907	-0.2987	424.9296	5253.0000	5.0000	50.0000	3.0000
5283.3197	4283.3197	4.3024	1.2858	-0.2775	439.0035	5408.0000	15.0000	30.0000	3.0000
5321.6811	4321.6811	4.3409	1.2827	-0.2562	452.5850	5380.0000	10.0000	30.0000	3.0000
5133.3698	4133.3698	4.1519	1.2530	-0.3607	476.6955	8182.0000	5.0000	30.0000	5.0000
4947.6690	3947.6690	3.9656	1.2528	-0.2487	409.3544	8086.0000	10.0000	30.0000	5.0000
4920.8117	3920.8117	3.9386	1.2497	-0.2948	390.1010	7878.0000	5.0000	50.0000	5.0000
5229.9739	4229.9739	4.2489	1.2314	-0.2372	348.5837	5203.0000	15.0000	50.0000	3.0000

Capitulation Parameter Robustness - Top 10 by Sharpe

final_value	profit	total_return	sharpe	max_drawdown	turnover_multiple	num_orders	max_return	min_volume_shock	n_winners
3792.8157	2792.8157	2.7928	1.2593	-0.1753	296.5980	479.0000	-0.0300	2.5000	3.0000
3861.9851	2861.9851	2.8620	1.2493	-0.1841	295.0743	297.0000	-0.0300	2.5000	1.0000
3779.4085	2779.4085	2.7794	1.2388	-0.1743	296.7234	406.0000	-0.0300	2.5000	2.0000
3121.7548	2121.7548	2.1218	1.1277	-0.1554	209.9036	332.0000	-0.0700	2.5000	3.0000
3110.6002	2110.6002	2.1106	1.1235	-0.1704	158.9610	258.0000	-0.0500	3.0000	3.0000
3231.7733	2231.7733	2.2318	1.1112	-0.1679	238.5852	399.0000	-0.0500	2.5000	3.0000
3139.2171	2139.2171	2.1392	1.1109	-0.1953	208.9679	210.0000	-0.0700	2.5000	1.0000
3118.4109	2118.4109	2.1184	1.1054	-0.1813	158.7386	216.0000	-0.0500	3.0000	2.0000
3304.6730	2304.6730	2.3047	1.0986	-0.2110	239.0817	239.0000	-0.0500	2.5000	1.0000
3065.2583	2065.2583	2.0653	1.0933	-0.1664	209.8363	280.0000	-0.0700	2.5000	2.0000

Parameter Robustness Interpretation

- The momentum strategy remains profitable across many nearby fast/slow moving-average settings.
- The original 10/30/3 configuration is not an isolated lucky parameter choice.
- Several alternative configurations, such as 15/30/5 and 10/50/2, also produce Sharpe ratios above 1.
- The capitulation strategy is more sensitive to parameter choices, especially the volume-shock threshold.
- Stricter volume confirmation generally improves capitulation performance and reduces drawdown.

BTC Regime Analysis

The strategies were also evaluated separately in BTC bull and bear regimes. A BTC bull regime is defined as a positive rolling 30-day BTC return, while a BTC bear regime is defined as a non-positive rolling 30-day BTC return.

BTC Regime Counts

regime	num_days
BTC bull regime	1011.0000
BTC bear regime	915.0000

BTC Regime Performance

strategy	regime	num_days	total_return	annualized_return	annualized_volatility	sharpe	max_drawdown
momentum	BTC bull regime	1011.0000	5.1999	0.9323	0.2312	2.9634	-0.1107
breadth	BTC bull regime	1011.0000	3.8229	0.7648	0.2339	2.5435	-0.0937
capitulation_daily	BTC bull regime	1011.0000	1.3493	0.3612	0.2639	1.2902	-0.1213
capitulation_selective	BTC bull regime	1011.0000	0.3842	0.1245	0.1327	0.9485	-0.0689
btc	BTC bull regime	1011.0000	168.3999	5.3783	0.5196	3.8298	-0.2138
momentum	BTC bear regime	915.0000	-0.5064	-0.2454	0.2373	-1.0681	-0.5685
breadth	BTC bear regime	915.0000	-0.4569	-0.2162	0.1409	-1.6574	-0.4923
capitulation_daily	BTC bear regime	915.0000	-0.3268	-0.1460	0.2657	-0.4614	-0.4364
capitulation_selective	BTC bear regime	915.0000	0.0949	0.0368	0.2283	0.2715	-0.3030
btc	BTC bear regime	915.0000	-0.9854	-0.8148	0.6101	-2.4494	-0.9863

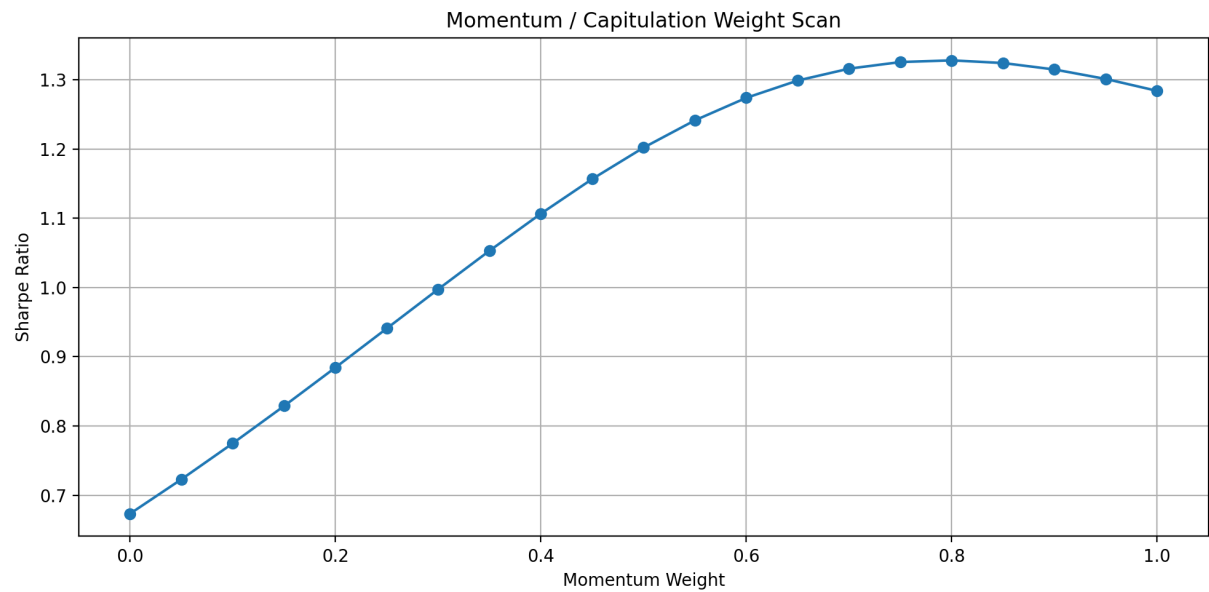
BTC Regime Analysis Interpretation

- Momentum performs best during BTC bull regimes, where trend persistence is strongest.
- Capitulation strategies are more defensive during BTC bear regimes.
- BTC buy-and-hold suffers the largest drawdowns in bear regimes.
- The selective capitulation strategy is the only tested strategy with positive Sharpe during BTC bear regimes.
- This supports combining trend-following and capitulation/reversal signals because they behave differently across market regimes.

Best Strategy Combinations

left_strategy	right_strategy	left_weight	right_weight	final_value	profit	sharpe	max_drawdown
momentum	capitulation_daily	0.8000	0.2000	4792.7776	3792.7776	1.3281	-0.1941
momentum	capitulation_selective	0.8000	0.2000	4265.7458	3265.7458	1.2995	-0.1850
breadth	capitulation_daily	0.8000	0.2000	3980.2393	2980.2393	1.3186	-0.2028
breadth	capitulation_selective	0.8000	0.2000	3536.6466	2536.6466	1.2875	-0.1850

Momentum + Capitulation Weight Scan



Best Combined Portfolio

The final portfolio combines the strongest standalone alpha, momentum, with the lower-correlation capitulation strategy. The optimal weight was selected by scanning portfolio weights from 0% to 100% in 5% increments and choosing the allocation with the highest Sharpe ratio. The best result was an 80% allocation to momentum and a 20% allocation to daily capitulation.

Best Combined Portfolio Metrics

weights	final_value	profit	sharpe	max_drawdown
{'momentum': 0.8, 'capitulation_daily': 0.2}	4792.7776	3792.7776	1.3281	-0.1941

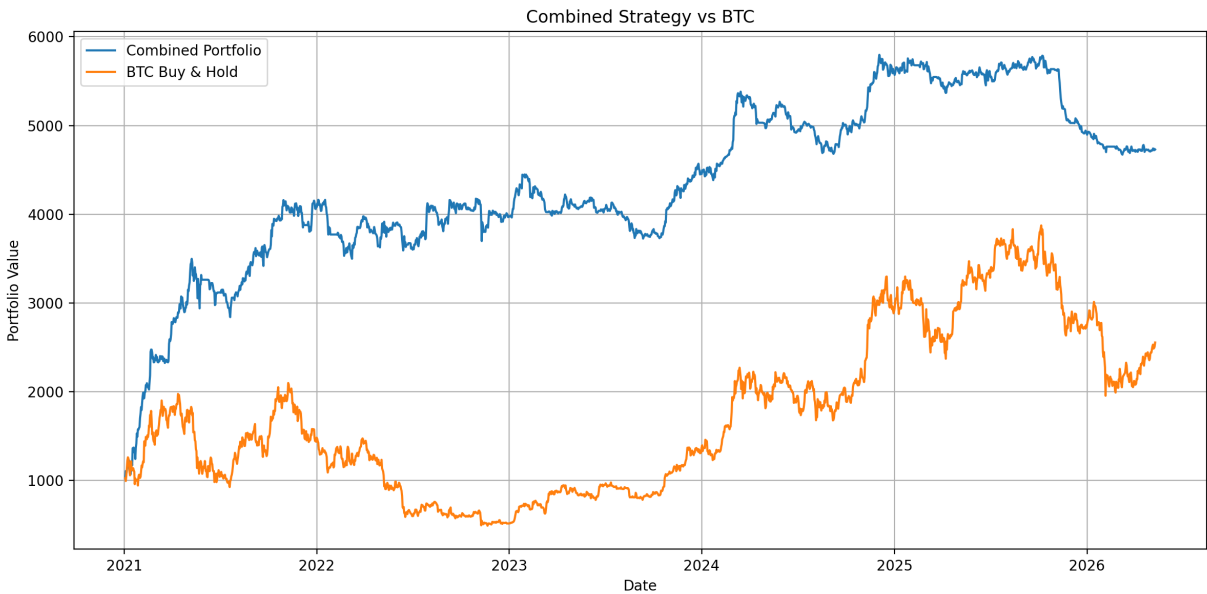
Best Combined Portfolio Alpha Report

alpha_daily	alpha_annual	beta_vs_btc	r_squared	information_ratio	alpha_t_stat	alpha_p_value
0.0009	0.3263	-0.0214	0.0026	1.3509	3.1233	0.0018

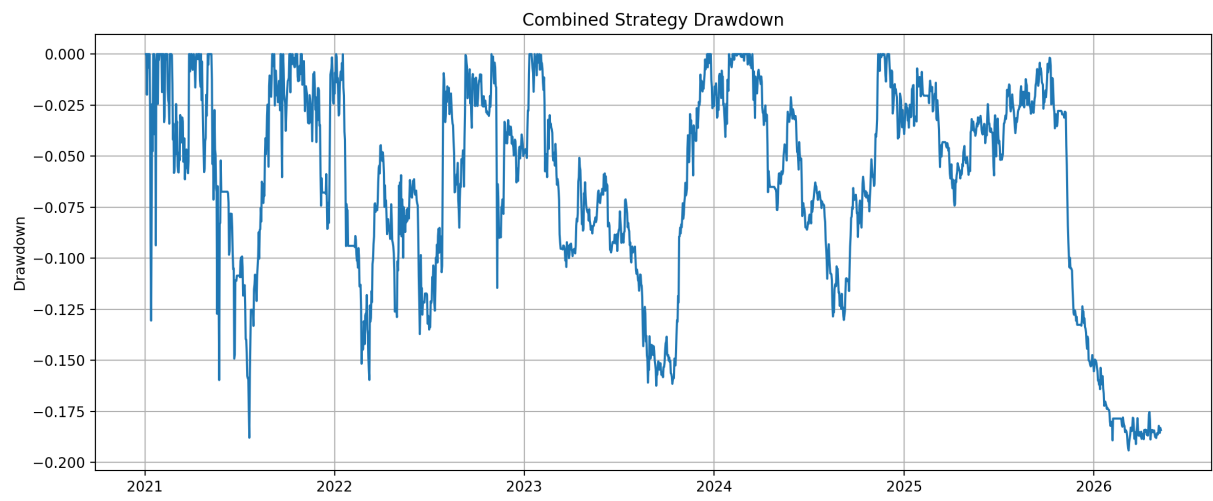
Best Combined Portfolio Benchmark Summary

asset	total_return	annual_return	annual_volatility	sharpe	max_drawdown
strategy	3.7296	0.3368	0.2419	1.3187	-0.1941
benchmark	1.5548	0.1915	0.5809	0.5918	-0.7663

Best Combined Portfolio vs BTC



Best Combined Portfolio Drawdown



Final Portfolio: Alpha Portfolio + BTC

After constructing the best alpha portfolio from momentum and capitulation signals, I tested whether adding a small BTC buy-and-hold allocation could improve the overall risk-adjusted return. Since the alpha portfolio had low beta to BTC, BTC exposure may add a complementary return stream rather than simply duplicating the same risk.

The best allocation found by the weight scan was 85% alpha portfolio and 15% BTC. Because the alpha portfolio itself consists of 80% momentum and 20% daily capitulation, the final effective allocation is approximately:

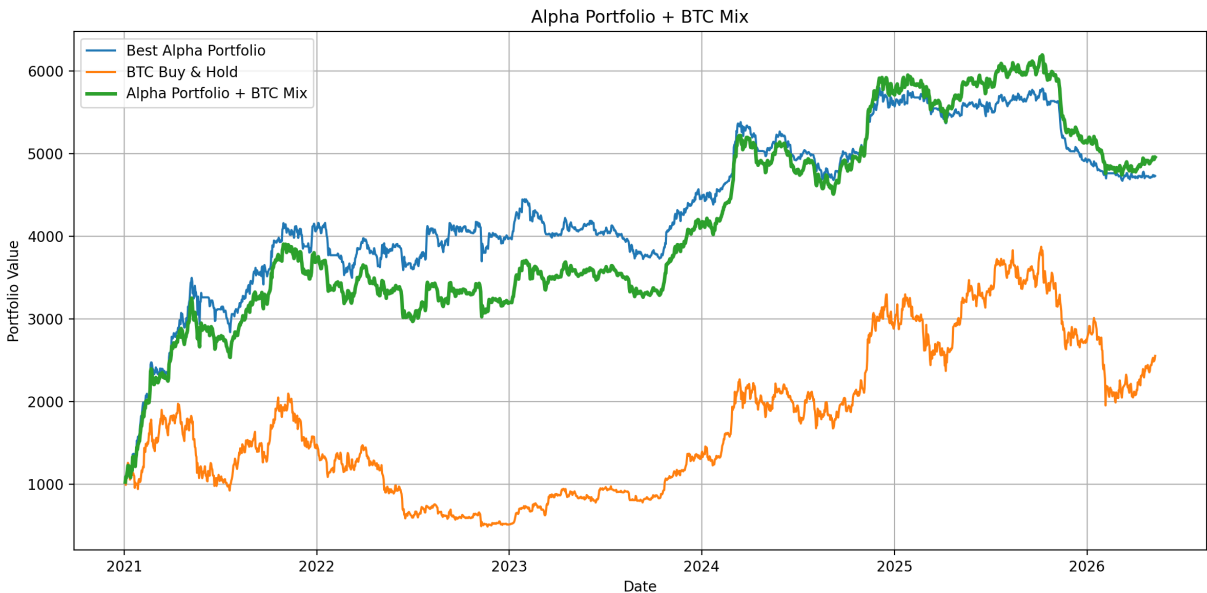
Component	Effective allocation
Momentum alpha	68%
Daily capitulation alpha	17%
BTC buy-and-hold	15%

Alpha Portfolio + BTC Best Mix

combined_strategy_weight	btc_weight	final_value	profit	sharpe	max_drawdown
0.8500	0.1500	4955.3689	3955.3689	1.4725	-0.2396

The final portfolio achieved a Sharpe ratio of approximately 1.47, compared with about 1.33 for the alpha-only portfolio. This suggests that a small BTC allocation improved portfolio-level diversification and risk-adjusted performance, despite BTC having much larger drawdowns as a standalone investment.

Alpha Portfolio + BTC Equity Curve



Key Findings

The dynamic momentum strategy delivered the strongest standalone performance, while the daily capitulation strategy provided a lower-correlation return stream. Combining 80% momentum and 20% daily capitulation improved portfolio Sharpe and reduced drawdown relative to standalone momentum. A final overlay that allocated 85% to this alpha portfolio and 15% to BTC further improved Sharpe to approximately 1.47. The resulting effective portfolio allocation was approximately 68% momentum alpha, 17% capitulation alpha, and 15% BTC buy-and-hold.

Limitations

This is a research backtest, not a production trading system. Limitations include possible survivorship bias in the candidate universe, simplified execution assumptions, absence of order book data, no funding-rate data, and limited out-of-sample validation.

Future Work

Future improvements include walk-forward validation, parameter sensitivity analysis, transaction-cost stress tests, volatility targeting, funding-rate integration, and live paper trading.